

Project documentation

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Introduction

This project is about creating a desktop application that would be connected to our PostgreSQL database and enable basic data operations.

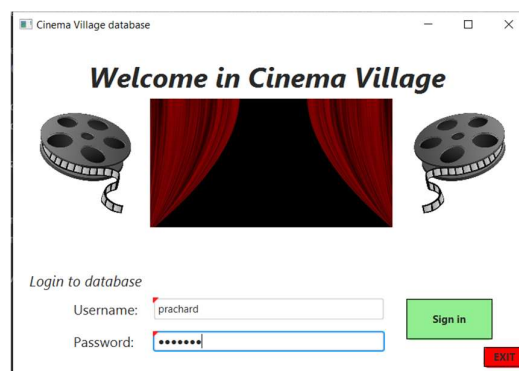
Project objectives to be achieved

- Creation of a desktop application program in java that connects the PostgreSQL database
- Running the application via the command line
- Creating a user role that will be able to access the database in PostgreSQL
- Authentication of the user who logs in to the database through desktop application
- Display basic data and detailed data(JOIN) from the PostgreSQL database in a desktop application
- Ability to use DML operations (create, update, select, delete) over data in the desktop application
- Filter data using the specified criteria
- Simulate SQL injections attacks
- Database backup
- Writing of events to the log file

Description of how the project works

First of all, there is a command ***mvn clean install***, that compiles the program and creates an executable jar file. We will then run our desktop application using the command ***java -jar target/Database_Cinema-1.0.0.jar***

The program will display a login form in our PostgreSQL database specifically to the CinemaVillage database. In order to continue, we must enter the correct login data consisting of username and password.



If we enter the wrong credentials, a display window will pop up informing that the login was unsuccessful.

If we enter the correct credentials, a display window will pop up informing that the login was successful. After that we will see selection menu, where we are able choose the following operations:

MENU

Please choose an operation:

Information about cinema branch

Update films

Injections SQL attacks

Project information

Partners

1. Information about cinema branch

When we pick the first option, the program will display a form window with tables and data. There is a main table of cinemas and 3 secondary tables that show us additional information related to the cinema (address, opening hours, other information). There is also an all address table, which uses the findALL function to show all addresses of cinemas and employees from our database.

To view more information, we have to click on a specific entry in the main table of cinemas.

Unfortunately, the display of additional information does not correspond to the entries in PostgreSQL databases.

The screenshot shows a window titled "Welcome to the menu" with a menu bar (File, Help) and a toolbar (Back to menu, Detail info, Cinema Village, Exit). The main table, "Cinema Village", lists 13 cinemas with columns: id_cinema, cinema_name, email, free_parking, and wheel_chair. The "Detail info" button is highlighted. To the right, three secondary tables are displayed: "Address" (id_address, city, street, house_number, zip_code), "Other information" (total_capacity, projections, count_of_movie_theater), and "Opening hours" (id_open_hours, opening, closing). The "All Address" table at the bottom shows a list of addresses.

id_cinema	cinema_name	email	free_parking	wheel_chair
1	OC Olšpys Bmo	olshpysbr@cinemavilla...	t	f
2	Faram Liberec	faramlib@cinemavilla...	t	t
3	Nová Klat	novasklat@cinemavilla...	t	f
4	Ořysm Palac Pardubice	orshpml@cinemavilla...	t	t
5	Přem Plac	plasppl@cinemavilla...	t	t
6	Ořysm Flora	orshpml@cinemavilla...	t	t
7	Eastfield Chodov	chodovpr@cinemavilla...	f	t
8	OC Leton	letonopr@cinemavilla...	f	f
9	OC Novy Zmochov	zmochovpr@cinemavilla...	f	t
10	Praha Slovansky barak	barakpr@cinemavilla...	f	t
11	Metropole Sltin	sltinpr@cinemavilla...	f	t
12	Faram Usti nad Labem	faramunl@cinemavilla...	t	f
13	OC Maly Spalcek	spalcekpr@cinemavilla...	t	t

id_address	city	street	house_number	zip_code
1	Praha	Na prikope	859/22	

total_capacity	projections	count_of_movie_theater
700	2D	14

id_open_hours	opening	closing
3	10:30:00	23:59:00

id_address	city	street	house_number	zip_code
1	Praha	Revnicka	121/1	15521
2	Praha	Vinohradska	2828/151	13000
3	Praha	Na prikope	859/22	

That is why there is an option to use the Detail info button, which will correctly display all additional information in the text fields

File Help

Back to menu

EXIT

Id cinema:

Cinemas name:

Email:

Free parking:

Wheelchair access:

City:

Street:

House number:

Zip code:

Total capacity:

Projections:

Count of movie theater:

Opening:

Closing:

2. Update films

The second option in Menu is CRUD operations on the movies table. The program will display a form window with records of movies from the database. We can manage some simple operations using buttons.

Back to menu

List of films

id_film ▲	films_name	footage_minutes	premiere
1	The Shawshank Redemption	142	
2	The Godfather	175	1972-03-24
3	The Dark Knight	152	
4	The Godfather Part II	202	
5	12 Angry Men	96	
6	Schindlers List	195	
7	The Lord of the Rings: The Return of the King	201	2002-01-10
8	Pulp Fiction	154	
9	The Lord of the Rings: The Fellowship of the Ring	178	
10	The Good, the Bad and the Ugly	178	
11	Forrest Gump	142	1994-10-20
12	Fight Club	139	
13	Inception	148	
14	The Lord of the Rings: The Two Towers	179	
15	Star Wars: Episode V - The Empire Strikes Back	124	1997-04-10

type footage in minutes

Delete film

Create film

Refresh

Edit film

EXIT

Delete - To delete movies from our database, we press - delete movie button. After that, form window pop up where we are able to enter the id of the movie, that we want to delete.

Create - To create a new movie, we will use the button - create film. Second step is fill the name of the movie, the length of the movie in minutes, or we can also enter the premiere date in the format (yyyy-mm-dd)

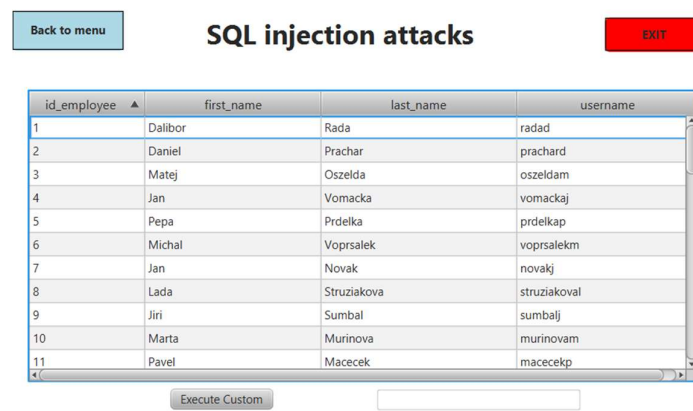
Update - To update the record, we must first click on the given movie that we want to change. Then click on the edit movie button and update the parameters that we want to change. We can change: movie id, movie title, length in minutes and premiere

Refresh - We can update the changes by using the refresh button

Filter - We can filter records using the criteria: movie length

3. Injections SQL attacks

Third option from Menu, will display us a table of employees of our cinemas. SQL attacks should have been simulated over this table, but this part is not implemented in the program



id_employee	first_name	last_name	username
1	Dalibor	Rada	radad
2	Daniel	Prachar	prachard
3	Matej	Oszelda	oszeldam
4	Jan	Vomacka	vomackaj
5	Pepa	Prdelka	prdelkap
6	Michal	Voprsalek	voprsalekm
7	Jan	Novak	novakj
8	Lada	Struziakova	struziakoval
9	Jiri	Sumbal	sumbalj
10	Marta	Murinova	murinovam
11	Pavel	Macecek	macecekp

4. Project information

Fourth option show us basic information about the project - author, name, creation date, license, etc

5. Partners

Five option display the icons of applications, that were used in my project and logos of companies, that create the applications.

Other functions

Exit - We are able to end the program at any time by use the exit button.

A step back - In each form window, we have the option to return to the previous form window using a back to menu button

Recording of errors and events in the log file

Backing up a PostgreSQL database using a script

Programs

The whole project was made in the INTELLIJ IDEA program environment. It is wrote in java language and maven. For the graphic part of the form windows i worked with the Scene Builder program and the java fx language. The database was created in the postgresSQL application.